

DS 2500 *Fall* 2025 Lab Exercise 07

Deadline : 10/27/2025 11:59 PM EST

Late Deadline: 11/03/2025 11:59 PM EST with 50 points off

Total Points: 200 pts

Grading:

- This lab assignment is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

Submission Instructions:

- Submit the Python file (`.py`) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

Additional Instructions:

- For Problems 1 and 2, you are welcome to use external packages to test whether your solution works. However, you may not use any external packages within your function implementations.

Disclosure: We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

Problem 1 (50 pts)

- Write a function called `zscore` to compute the z-score for a list of numbers.
 - If an empty list is passed, the function should return `None`.

Problem 2 (50 pts)

- Write a function called `min_max_scaling` to compute the Min-Max scaling for a list of numbers, scaling the data to a range between 0 and 1.
 - If an empty list is passed, the function should return `None`.
 - Assume that the list passed to the function will always have different values for the minimum and maximum.

Problem 3 (50 pts)

```
import pandas as pd

df = pd.DataFrame({
    'A': [1, 2, 3, 4, 5],
    'B': [4, None, None, None, None],
    'C': ["High", "Medium", "Medium", "High", "Low"]
})
```

- Create the `df` dataframe above.
- Use `df` to remove all columns that contain any missing values and store the result in a new dataframe named `df_col_cleaned`.
- Use `df` to remove all rows that contain any missing values and store the result in a new dataframe named `df_row_cleaned`.
- Examine `df_col_cleaned` and `df_row_cleaned` to understand how the data has changed after each operation.
- Hint: Use the Pandas documentations linked below to help you complete this task:
 - <https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.drop.html>
 - <https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.dropna.html>

Problem 4 (50 pts)

- Using `df_col_cleaned`, create a one-hot encoding of the categorical column.
- Store the dataframe with one-hot encodings to `df_encoded`.
- Examine `df_encoded` to verify that the one-hot encoding was applied correctly.
- Hint: Use the Pandas documentation linked below to help you complete this task:
 - https://pandas.pydata.org/docs/reference/api/pandas.get_dummies.html

Extra Problem 1 (0 pt; not graded)

1. True or False: Bar plot is useful for checking outliers.
2. True or False: Mean depends on the distribution of data.
3. True or False: Median is the average of maximum and minimum values of the data.
4. True or False: Each bin in a histogram represents a single data point.
5. True or False: Scatter plot is useful for checking the relationship between two continuous variables.